

Areg Hovumyan

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EDUCATION

California State Polytechnic University, Pomona

Pomona, CA

Bachelor of Science in Computer Engineering — GPA: 3.6/4.0

Expected Dec. 2027

- **Relevant Coursework:** Digital Logic Design, Microcontrollers & Embedded Systems, Circuit Analysis, Signals & Systems, Data Structures & Algorithms

RELEVANT EXPERIENCE

NASA L'SPACE Mission Concept

May 2026 – Present

Electrical Member – Robotic Lunar Lander Concept

Remote

- **Presented the C&DH progress** to NASA L'SPACE staff to evaluate and consider for an actual mission.
- **Designed the power subsystem architecture:** solar battery selection and power budget allocation across the subsystems.
- Developed MCUs, regulators, and AC-DC converters by calculating radiation tolerance and power draw.
- Built **Command & Data Handling (C&DH) architecture**, including communication between the Earth and the Moon.
- Worked alongside the science, thermal, and mechanical subteams as an electrical subsystem member.

PROJECT EXPERIENCE

Software Engineering Team Lead: Autonomous Drone

Jun. 2025 – Present

Sponsored by Lockheed Martin

Pomona, CA

- Collaborated with the Electrical and Payload teams to ensure safe and successful **flight deployments**.
- Presented **Lockheed Martin engineers** with the **system design** during a **Preliminary Design Review (PDR)**.
- Developed an **object recognition pipeline** on a **Jetson**, achieving **3x quicker detection** and increased accuracy to **99%**.
- Implemented **multithreading in C++/Python** using **lock-protected shared state**.
- **Improved runtime performance and cross-thread data consistency by 1.5x**.

Embedded Systems Team Lead: Autonomous Ground Vehicle

Jan. 2026 – Present

Northrop Grumman Collaboration Project

Pomona, CA

- Presented **telemetry design** to **Northrop Grumman engineers** during a **Preliminary Design Review (PDR)**.
- Created **reusable device drivers** and **register-level C code** for microcontroller peripherals.
- Used **logic analyzers** to debug problems that cross **electrical, firmware, and software borders**.
- Created and evaluated **multi-agent autonomy behaviors** in **actual flight conditions**.

Software Engineer: Autonomous Vehicle Lab

Apr. 2026 – Present

University Research Lab

Pomona, CA

- Implemented **Teensy, STM32**, and other microcontrollers for controlling embedded vehicle systems.
- Designed a **manually controllable and autonomous rover** for a competition.
- Developed **CI/CD pipelines** for the team to push safer code to production.
- Created **unit tests using GoogleTest and pytest** to improve system reliability.
- Utilized AI agents to refactor, rewrite, improve, and test the software.

Full-Stack Software Engineer/Founder

May 2025 – Dec. 2025

Web Development Company Founder

Remote

- Led a team of 10 engineers to build full-stack websites for various clients.
- Created websites that include chatbots, interactive interfaces, futuristic designs, and proper databases.
- Created safety guardrails, debugging pipelines, and monitoring tools to improve the reliability of all the products.
- Hired engineers from different disciplines such as systems, web, HR, and DevOps.

TECHNICAL SKILLS

Languages: C, C++, Python, JavaScript, TypeScript, Verilog, VHDL, ARM Assembly, Bash, SQL

Embedded: STM32 (bare-metal, register-level), ESP32, UART, SPI, I2C, DMA, interrupts, timers/PWM, logic analyzers, schematics

Robotics / CV: ROS2 (Humble/Iron), MAVLink, Gazebo, RPLIDAR, Jetson Orin Nano, YOLOv8/v11, PyTorch, TensorRT, ONNX, OpenCV, Roboflow

Tools: Linux, Git, GitHub Actions, Docker, Kubernetes, FastAPI, PostgreSQL, MongoDB, pytest, gtest